Don't just read about data structures. Build them, test them and time them. Every structure in this book is a working class you write yourself, checked against Python's built-ins on thousands of random inputs and measured with real benchmarks, so you see Big O happen on your own screen.

Build arrays, linked lists, hash tables, heaps, search trees, an AVL tree, a trie, union-find and graphs; implement eight sorts, Dijkstra, Kruskal and more. Then master the interview patterns: two pointers, sliding window, prefix sums, BFS and DFS, dynamic programming, greedy and backtracking.

Every listing was run on Python 3.12 to 3.15. Every chapter has real-world uses and exercises with tested solutions.

Aadhavan Muthurengan is an automation and AI professional with more than 15 years of experience in the automotive industry, where Aadhavan has built in-house tools that transformed product development and validation.

DATA STRUCTURES AND ALGORITHMS IN PYTHON, HANDS-ON

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Build Every Structure Yourself, Benchmark It, and Master the Interview Patterns

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